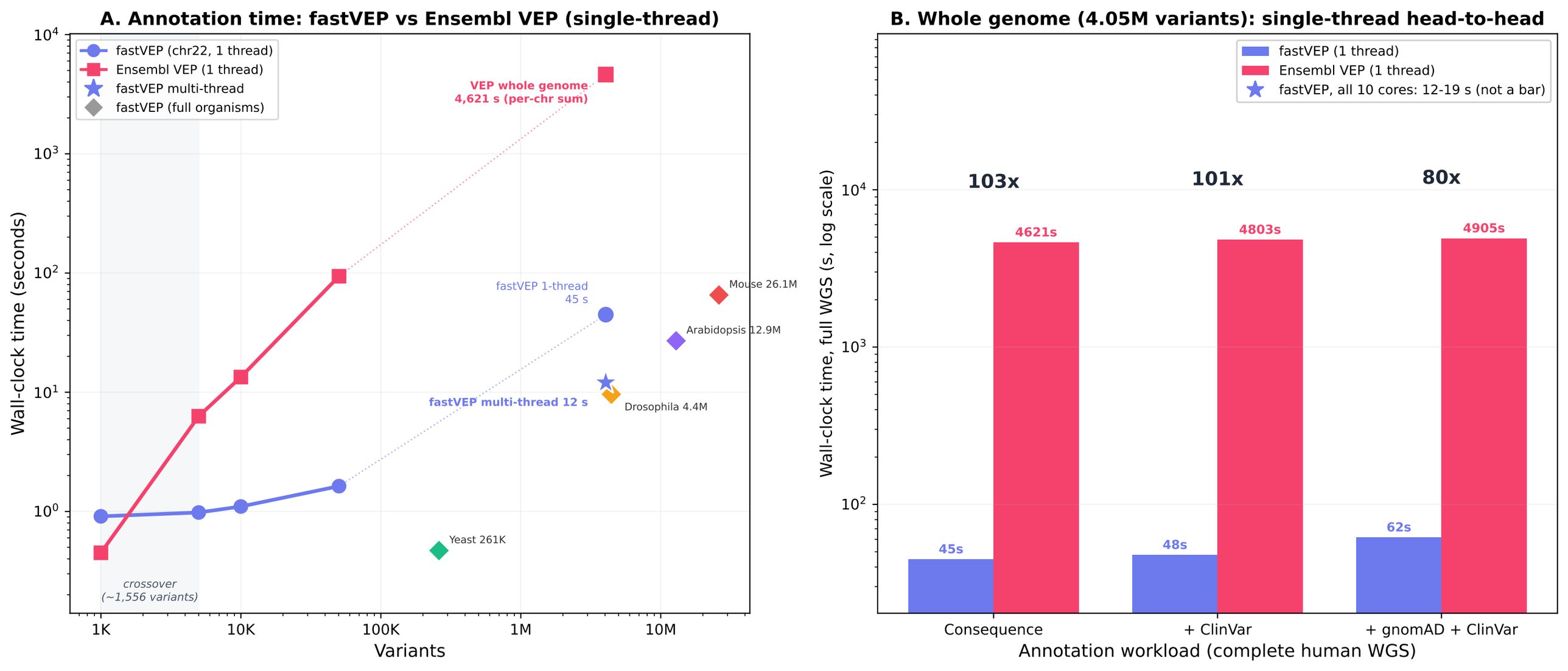
A. Annotation time: fastVEP vs Ensembl VEP (single-thread)
fastVEP (chr22, 1 thread)
Ensembl VEP (1 thread)
fastVEP multi-thread
fastVEP (full organisms)
VEP whole genome
4,621 s (per-chr sum)
fastVEP 1-thread
45 s
fastVEP multi-thread 12 s
Mouse 26.1M
Arabidopsis 12.9M
Drosophila 4.4M
Yeast 261K
crossover
(~1,556 variants)
Wall-clock time (seconds)
Variants
1K
10K
100K
1M
10M
B. Whole genome (4.05M variants): single-thread head-to-head
fastVEP (1 thread)
Ensembl VEP (1 thread)
fastVEP, all 10 cores: 12-19 s (not a bar)
103x
101x
80x
4621s
4803s
4905s
45s
48s
62s
Wall-clock time, full WGS (s, log scale)
Consequence
+ ClinVar
+ gnomAD + ClinVar
Annotation workload (complete human WGS)